MIL-G-51025H
18 November 1968
SUPERSEDING
MIL-G-0051025G(MU)
27 June 1968
MIL-G-51025F
5 August 1966

MILITARY SPECIFICATION

GUN, PORTABLE RIOT CONTROL

AGENT DISPERSER, M9

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of agent disperser gun.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Steel, Strapping, Flat.
TT-W-572 - Wood Preservative.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-P-116 - Preservation, Methods of.

FSC 1040

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL116-4-16 - Gun, Portable Riot Control Agent Disperser, M9.
IDL116-4-16 - Gun, Portable Riot Control Agent Disperser, M9 - Inspection Data List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material and components.

3.1.1 Materials. All materials cited on Drawing DL116-4-16 and on the subsidiary drawings shall conform to the specifications thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the gun shall conform to the specifications and drawings listed on Drawing DL116-4-16 and subsidiary drawings.

3.2 Assembly. The gun shall be assembled as shown on Drawing D116-4-16.

3.3 Trigger functioning. There shall be full trigger travel when tested as specified in 4.4.4.1.

3.4 Leakage and functioning. The gun shall not leak and shall function without binding of the trigger assembly when tested as specified in 4.4.4.2.

3.5 Sleeve attachment. The sleeve (Drawing B116-4-27) shall not separate from the barrel assembly (Drawing C116-4-26) when subjected to a force of 1000 $\frac{+25}{-0}$ pounds when tested as specified in 4.4.4.3.

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3.6 Rubber tube, Drawing D116-4-16.

3.6.1 Tensile strength. The tube shall have a minimum average tensile strength of 1000 pounds per square inch (psi) when tested as specified in 4.4.4.4(a).

3.6.2 Elongation. The tube shall have a minimum average elongation of 360 percent when tested as specified in 4.4.4.4(a).

3.6.3 Breakage under tension. The tube shall not break when tested as specified in 4.4.4.4(b).

3.7 Serial number. Each gun shall be identified by a serial number as specified in the contract or order (see 6.2).

3.8 Preproduction. Prior to the start of regular production, a preproduction sample of guns shall be produced in accordance with this specification for examination and tests (see 4.3).

3.9 Workmanship. The gun shall be free of foreign matter (dirt, oil or grease) except where grease is required by drawings and damage such as dents, cracks or deformation that could affect assembly or functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction inspection

4.3.1 Sample. A preproduction sample of five guns shall be manufactured using the same methods, materials, equipment, processes and procedures as will be used during regular production. Any change in materials, methods, process or procedures will require a new preproduction sample.

4.3.2 Inspection procedure

4.3.2.1 For examination and nondestructive tests. The sample guns shall be examined and tested in accordance with the classification of defects (4.4.3.4) and 4.3.3. The dimensions of components shall be recorded. The AQL does not apply.

4.3.2.2 For destructive tests. After the above nondestructive tests, the rubber tubes shall be removed from the guns and tested in accordance with 4.4.4.4. The acceptance number is zero.

4.3.3 Trigger functioning test. Completely depress the trigger and then release it. Full trigger travel is indicated when the safety mechanism engages. Repeat this test a minimum of 100 times. After this test gun will be disassembled and spring examined. Cracked or broken springs shall be cause for rejection.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the guns of one interfix number, produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of components of any type shall be assembled into any one lot of guns. A lot of any type component shall consist of that type component manufactured at one plant, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample guns shall be examined and tested in accordance with the classification of defects and MIL-STD-105. The level A preparation for delivery shall be examined in accordance with 4.4.3.5.

4.4.3.2 For destructive tests. The sample tubes shall be tested in accordance with 4.4.4.4 using an AQL of 2.5 percent defective.

4.4.3.3 For critical defects. Each item in the lot shall be tested for the critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects. (See inspection data list IDL116-4-16 for inspection equipment required to perform the examinations specified in this section.)

(a) Gun assembly (Drawing D116-4-16).

Categories	Defects	Acceptance standards
Critical:		
1	Trigger functioning	4.4.4.1
2	Leakage and functioning	4.4.4.2
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Serial number missing	
103	Spring and trigger grease missing	
104	Workmanship (3.9)	

(b) Tube, rubber (Drawing D116-4-16).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Tube cracked or split
102	Length incorrect
103	Outside diameter incorrect
104	Wall thickness incorrect

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4.4.3.4 Classification of defects (continued).

(c) Collar (Drawing B116-4-33).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Tapped hole incorrect
102		Diameter large bore incorrect
103		Eccentricity excessive
104		Diameter small bore incorrect
105		Workmanship (3.9)
Minor:	AQL 4.0 percent defective	
201		Chamfer missing
202		Protective finish missing or incorrect

(d) Sleeve (Drawing B116-4-32)

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Taper incorrect
102		Diameter large end taper incorrect
103		Outside diameter incorrect
Minor:	AQL 4.0 percent defective	
201		Chamfer missing
202		Protective finish missing or incorrect

(e) Spring, trigger (Drawing B116-4-25).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Profile after heat treating incorrect
102		Protective finish missing or incorrect

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4.4.3.4 Classification of defects (continued).

(f) Connector (Drawing B116-4-23).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Diameter (.530 max) incorrect
102		Diameter (.590) incorrect
103		Outside diameter incorrect
104		Taper (5°) missing
105		Sharp edges
Minor:	AQL 4.0 percent defective	
201		Protective finish missing or incorrect

(g) Trigger (Drawing D116-4-24).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Outside diameter incorrect
102		Eccentricity excessive
103		Location of square hole incorrect
104		Height of square hole incorrect
105		Depth of groove (.670) incorrect
106		Spring slot width incorrect
107		Location of spring slot incorrect
108		Outside radius (max) spring retaining pads incorrect
109		Surface 39° 30' incorrect
110		Sharp edges or burrs in square hole
111		Finish incorrect
112		Location and diameter (.139) hole incorrect
113		Overall width incorrect
114		Dimension (5/8 ∠ 1/64) incorrect
115		Width (3/8) incorrect
Minor:	AQL 4.0 percent defective	
201		Protective finish missing or incorrect

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4.4.3.4 Classification of defects (continued).

(h) Plate, cover (Drawing C116-4-29).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Location and diameter of holes incorrect
102		Angle and width of tab incorrect
103		Boss not flush
104		Finish (32) incorrect
105		Length of tab (max) incorrect
106		Location of tab incorrect
Minor:	AQL 4.0 percent defective	
201		Protective finish missing or incorrect

(i) Safety, trigger (Drawing C116-4-28).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Component missing or incorrectly assembled
102		Spring profile incorrect (Drawing B116-4-18)
103		Spring thickness incorrect (Drawing B116-4-18)
104		Rockwell hardness incorrect (Drawing B116-4-18)
105		Diameter (.125) hole incorrect (Drawing C116-4-30)
Minor:	AQL 4.0 percent defective	
201		Protective finish missing or incorrect

(j) Barrel assembly (Drawing C116-4-26).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL 1.0 percent defective		
101		Component missing or incorrectly assembled	
102		Sleeve brazing incorrect	4.4.4.3
103		Taper incorrect	
104		Diameter (.805) incorrect	

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4.4.3.4 Classification of defects (continued).

(j) Barrel assembly (Drawing C116-4-26).

Categories	Defects
Major:	
105	Thread incorrect (Drawing B116-4-31)
106	Undercut missing (Drawing B116-4-31)
107	Diameter large end of taper incorrect
108	Width of tab incorrect (Drawing B116-4-27)
109	Nut binding
Minor:	AQL 4.0 percent defective
201	Chamfers missing
202	Knurl missing (Drawing B116-4-31)
203	Protective finish missing or incorrect
204	Workmanship (3.9)

(k) Body, valve (Drawing D116-4-17).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Insert missing or incorrectly assembled
102	External thread incorrect
103	Internal thread incorrect
104	Tapped holes incorrect or location incorrect
105	Diameter (.810) incorrect
106	Width and location of slot (7/16) incorrect
107	Depth (1.626) incorrect
108	Finish (32) incorrect
109	Angle and width of slot (.656) incorrect
110	Diameter (1.500) incorrect
111	Location and height of square hole incorrect
112	Sharp edges on square hole
113	Position of slot (.250) incorrect
114	Width of slot (.250) incorrect
Minor:	AQL 4.0 percent defective
201	Insert 120° angle missing
202	Barrel end 60° angle missing
203	Chamfers missing
204	Protective finish missing or incorrect

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4.4.3.5 List of defects.

(a) Preparation for delivery (section 5).

1. Component missing
2. Fiberboard cells missing
3. Containers incorrect
4. Container closure incorrect
5. Container damaged (broken, punctured or crushed)
6. Quantity per container incorrect
7. Marking missing, illegible or incorrect

4.4.4 Tests. Tests shall be conducted as follows: (See inspection data list IDL116-4-16 for inspection equipment required to perform the tests specified in this section.)

4.4.4.1 Trigger functioning. Completely depress the trigger and then release it. Full trigger travel is indicated when the safety mechanism engages. Repeat this test a minimum of 15 times.

4.4.4.2 Leakage and functioning. Connect the gun to a 75 ± 5 pounds per square inch gage (psig) source of air. Immerse the forward end of the barrel in water and observe for bubbles. Using only enough force to take up any slack in the safety mechanism, attempt to depress the trigger and observe for bubbles. Remove the gun barrel from the water and actuate the trigger in a few short bursts. Note the smoothness of the trigger movement. A rough or jerky movement indicates binding of trigger assembly, absence of lubricant, or a twisted rubber tube. Completely depress and release the trigger. The safety mechanism should engage. Immerse the entire gun in water and observe for bubbles for a minimum of 30 seconds. Repeat the leakage test a minimum of two times. The presence of air bubbles during any part of the test indicates leakage. Disconnect the gun from the source of air and disassemble for packaging as follows:

(1) Remove collar and sleeve from nozzle end.
(2) Depress trigger and remove rear connector with rubber tube.
(3) Unscrew lock nut holding gun barrel to gun body, separating into two sections.
(4) Dry parts thoroughly prior to packaging.
(5) Package in accordance with 5.1 at completion of tests.

4.4.4.3 Sleeve attachment. Secure the barrel assembly in a fixture. At a rate of 1 inch per minute, in a direction away from the handle and parallel to the barrel, apply force to the sleeve until the specified force is applied. If the sleeve becomes separated or loose, the barrel assembly shall be rejected.

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4.4.4.4 Rubber tube. Measure the outside diameter and wall thickness in four places on each tube. Use the average values and calculate the cross-sectional area of the tube. Cut the tube into four 6-inch specimens and place a 1-inch die mark at the center of each specimen.

(a) Tensile strength and elongation. Place a tube specimen in a horizontal tensile testing machine. With a jaw travel speed of 12 inches per minute, stretch the specimen to failure. Compute the tensile strength by dividing the breaking strength by the cross-sectional area. Determine percent elongation by measuring the die mark at the instant the specimen failed and determining the percent change from the original 1-inch mark. Perform this test on three of the four six-inch specimens and calculate the average value for the three specimens.

(b) Breakage under tension. Elongate the other 6-inch specimen 100 percent (as measured from the 1-inch die mark) and retain the specimen in that condition for minimum period of 24 hours.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. The disassembled gun (see 4.4.4.2) shall be packaged together in accordance with method IA-14 of MIL-P-116. The inner box shall conform to style RSC, W5c or W6c of PPP-B-636 and have inside dimensions of 6-1/4 by 3-3/8 by 17-1/8 inches, length, width and depth respectively. The outer box shall conform to style RSC, W5c or W6c of PPP-B-636 and have inside dimensions of 6-3/4 by 3-3/4 by 17-3/4 inches, length, width and depth respectively. The gun shall be placed into the inner box in accordance with Drawings D116-4-37, C116-4-38 and C116-4-39. The box shall be closed by taping in accordance with the requirements for weather-resistant boxes in the appendix to PPP-B-636. Included air volume shall be kept to a minimum in the sealed barrier bag. Filler pads of weather-resistant class fiberboard shall be included, as necessary, to restrict free movement of the box contents.

5.1.2 Level C. The disassembled gun (see 4.4.4.2) shall be packaged in a fiberboard box in accordance with Drawing D116-4-37. The box shall have dimensions of 6-1/4 by 3-3/8 by 17-1/8 inches, length, width and depth respectively and may be fabricated from either domestic class or weather-resistant class fiberboard, including filler pads as necessary to restrict free movement of the box contents. The box shall be closed by taping in accordance with the requirements for weather-resistant boxes in the appendix of PPP-B-636.

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5.2 Packing.

5.2.1 Level A. The guns, packaged in accordance with 5.1.1, shall be packed, 10 per box, in a wood box conforming to style 4, class 2, for type 2 load of PPP-B-621 having inside dimensions of 20-1/2 by 18-1/2 by 14-1/4 inches, length, width and depth respectively. The box or wood components shall be treated by immersing in wood preservative conforming to type II, composition A of TT-W-572, for a minimum period of three minutes. Boxes shall not be used for at least 24 hours after treatment. Arrangement in the boxes shall be two tiers of five each. Filler pads (see 5.1.1) shall be used as necessary to restrict free movement of the packages. The box shall be closed and strapped in accordance with the appendix to PPP-B-621. Strapping shall conform to class B of QQ-S-781. Weight of the contents of box is approximately 58 pounds.

5.2.2 Level B. The guns, packaged in accordance with 5.1.1 shall be packed 10 per box in fiberboard boxes conforming to V3c of PPP-B-636, having inside dimensions of 18-1/2 by 14-1/4 by 20-1/2 inches length, width and depth respectively. Filler pads of weather-resistant fiberboard conforming to PPP-F-320 shall be used to line all interior faces of the box, with additional pads to restrict free movement of the packages. Boxes shall be closed by gluing in accordance with requirements for weather-resistant boxes in the appendix of PPP-B-636. All seams and joints shall be sealed with tape as required for water-proof sealing slotted style boxes and reinforced by tape banding in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packing shall be in accordance with 5.2.2, except the box may be domestic class and sealing of the seams and joints and tape banding is not required.

5.3 Palletization. When specified in the contract or order (see 6.2), the packed guns shall be palletized and marked in accordance with Drawing D116-4-36. Arrangement shall be 2 by 2 by 2 boxes (8 boxes per pallet).

5.4 Marking. In conjunction with requirements stipulated in the contract or order (see 6.2), marking shall be as specified herein.

5.4.1 Package. Packages shall be marked in accordance with applicable packaging marking required by MIL-STD-129.

5.4.2 Packing. Exterior boxes (shipping containers) shall be marked in accordance with applicable requirements of Drawing D150-2-1.

5.4.3 Palletization. See 5.3.

5.5 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

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6. NOTES

6.1 Intended use. This item is intended for use in directing agent (in powder form) from riot control dispersers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packaging and packing required.
(c) Serial numbers for guns.
(d) Appropriate identification for the item for inclusion in marking.
(e) Appropriate address for inclusion in marking.
(f) Palletization when required (mandatory for level A packing).
(g) Preservative treatment of pallets (see 5.2.1).

Custodian:

Army - MU
Navy - MC
Air Force - 84

Review activity:

Army - SM
Navy - MC
Air Force - 84

Preparing activity:

Army - MU(EA)

Project No. 1040-0047

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 55 WHICH MAY BE USED

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